ODD LOG-PRIME MERTENS KERNELS AND CONDITIONAL DIRICHLET-SINGULARITY TRANSFER

Abstract. This paper establishes finite-range cancellation for the odd log-prime Mertens kernel

$$Y_t(X) = \sum_{k \le X} \mu(k) \sin\big(t \log(cp_k)\big), \tag{0.1}$$

where p_k is the k-th prime and $c > 0$ is fixed, and identifies the analytic conditions under which such cancellation would extend to a hypothetical asymptotic regime. The numerical part records suppression of $Y_t(X)/\sqrt{X}$ and of the intrinsic variance-normalized statistic

$$Z_t(X) = \frac{Y_t(X)}{\Big(\sum_{k \le X} \mu(k)^2 \sin^2(t \log(cp_k))\Big)^{1/2}} \tag{0.2}$$

up to primes $p_k \le 2.5 \cdot 10^8$. Three low-escape frequencies selected from the extended sweep, $t = 3.0437$, $t = 0.48243$, and $t = 7.8337$, exhibit strong cancellation on the internal variance scale; for the first two, trajectory-level random-sign tests show compression against null walks on the same squarefree support, not merely endpoint cancellation.

The analytic object attached to the kernel is the Dirichlet series

$$D_t(s) = \sum_{k \ge 1} \frac{\mu(k) \sin(t \log(cp_k))}{k^s}. \tag{0.3}$$

A direct Abelian argument shows that a square-root scale bound for $Y_t(X)$ forces analytic continuation of $D_t(s)$ into the corresponding half-plane. The singularity calculation then uses the Cipolla expansion of the indexed prime and a strong fractional transfer hypothesis for logarithmic Dirichlet symbols. This strong transfer hypothesis does not follow, at least trivially, from the Riemann Hypothesis: RH controls the global location of zeros, while the present hypothesis concerns finer local analytic stability of shifted reciprocal-zeta singularities under the exact prime-index symbol. Under this hypothesis, each simple zero ρ of ζ produces a non-removable branch of D_t at $s = \rho \pm it$, with leading coefficient proportional to $1/\zeta'(\rho)$. Consequently, the strong transfer hypothesis together with a square-root bound for the odd kernel gives a conditional implication to the Riemann Hypothesis for simple zeros.

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1. Introduction

Let μ denote the Moebius function and let p_k denote the k-th prime. This paper considers the one-parameter odd log-prime kernel

$$Y_t(X) = \sum_{k \le X} \mu(k) \sin\big(t \log(cp_k)\big), \tag{1.1}$$

where $c > 0$ is fixed. The corresponding complex kernel is

$$A_t(X) = \sum_{k \le X} \mu(k)(cp_k)^{it}, \tag{1.2}$$

and the odd kernel is its imaginary part:

$$Y_t(X) = \Im A_t(X). \tag{1.3}$$

The classical Mertens function is

$$M(X) = \sum_{n \le X} \mu(n). \tag{1.4}$$

It is tied directly to the reciprocal zeta function through

$$\sum_{n \ge 1} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}, \qquad \Re s > 1. \tag{1.5}$$

The Odlyzko–te Riele disproof of the Mertens conjecture uses this connection together with an explicit formula, smoothing, and lattice approximation of zero phases [7]. Standard background on ζ, Dirichlet series, Perron and Abel summation, and prime number theory can be found in [9, 4, 3, 6, 5].

The purpose here is to isolate the analogous analytic object for (1.1). The kernel contains two arithmetically simple ingredients: the Moebius coefficient $\mu(k)$ and the indexed prime p_k. The sine selects the odd part of the complex twist. This odd projection is numerically small in the computed range, and the analytic question is whether the associated Dirichlet series retains the same singularity mechanism as $1/\zeta$.

Remark 1.1. The statements below are separated by status. Lemmas and propositions are rigorous under their stated hypotheses. Formal transformations using fractional logarithmic differentiation are explicitly placed in heuristic environments. Numerical statements are recorded as experiments.

2. The odd log-prime kernel

Definition 2.1 (Odd log-prime Mertens kernel)**.** For fixed $c > 0$ and $t \in \mathbb{R}$, define

$$Y_t(X) = \sum_{k \le X} \mu(k) \sin\big(t \log(cp_k)\big). \tag{2.1}$$

The associated complex kernel is

$$A_t(X) = \sum_{k \le X} \mu(k)(cp_k)^{it}. \tag{2.2}$$

Since

$$\sin\big(t \log(cp_k)\big) = \frac{(cp_k)^{it} - (cp_k)^{-it}}{2i}, \tag{2.3}$$

the kernel can be written as

$$Y_t(X) = \frac{c^{it}}{2i} \sum_{k \le X} \mu(k) p_k^{it} - \frac{c^{-it}}{2i} \sum_{k \le X} \mu(k) p_k^{-it}. \tag{2.4}$$

Because $\mu(k)$ is real, this is precisely the odd part in the t variable:

$$Y_{-t}(X) = -Y_t(X). \tag{2.5}$$

Definition 2.2 (Dirichlet series of the odd kernel)**.** For $\Re s > 1$, define

$$D_t(s) = \sum_{k\geq 1} \frac{\mu(k)\sin(t\log(cp_k))}{k^s}. \tag{2.6}$$

Equivalently,

$$D_t(s) = \frac{c^{it}}{2i}\sum_{k\geq 1}\mu(k)p_k^{it}k^{-s} - \frac{c^{-it}}{2i}\sum_{k\geq 1}\mu(k)p_k^{-it}k^{-s}. \tag{2.7}$$

The series in (2.6) is absolutely convergent for $\Re s > 1$, since $|\mu(k)\sin(t\log(cp_k))| \leq 1$.

3. Abelian continuation from a square-root scale bound

The first step is the exact analogue of the classical Abelian passage from a bound on a summatory function to continuation of its Dirichlet series. The proof is standard; it is included because it is the logical bridge used in the reduction.

Lemma 3.1 (Abelian continuation)**.** *Let*

$$a_k = \mu(k)\sin\big(t\log(cp_k)\big), \tag{3.1}$$

and

$$A(X) = \sum_{k\leq X} a_k. \tag{3.2}$$

If, for some $\alpha \geq 0$ and every $\varepsilon > 0$,

$$A(X) = O_{\varepsilon,t,c}(X^{\alpha+\varepsilon}), \tag{3.3}$$

then

$$D_t(s) = \sum_{k\geq 1} a_k k^{-s} \tag{3.4}$$

extends holomorphically to the half-plane

$$\Re s > \alpha. \tag{3.5}$$

Proof. For $\Re s > 1$, partial summation gives

$$D_t(s) = s\int_1^\infty A(x)x^{-s-1}\,dx. \tag{3.6}$$

The hypothesis (3.3) implies convergence of the integral locally uniformly for $\Re s > \alpha + \varepsilon$. Since ε is arbitrary, this gives the stated continuation. □

Corollary 3.2 (Square-root scale continuation)**.** *If*

$$Y_t(X) = O_{\varepsilon,t,c}(X^{1/2+\varepsilon}) \tag{3.7}$$

for every $\varepsilon > 0$, then $D_t(s)$ extends holomorphically to

$$\Re s > \frac{1}{2}. \tag{3.8}$$

This is the same structural move used for the Mertens function, where $M(X) = O(X^{1/2+\varepsilon})$ would imply analytic continuation of $1/\zeta(s)$ into $\Re s > 1/2$ by (1.5).

4. Prime-index expansion and the principal zeta model

The prime-index input is the asymptotic expansion of the k-th prime. The first terms are

$$p_k = k\log k\left(1+\frac{\log\log k-1}{\log k}+O\left(\frac{(\log\log k)^2}{(\log k)^2}\right)\right), \tag{4.1}$$

with higher terms given by the Cipolla expansion; see [1, 2, 8]. For fixed t, (4.1) gives the formal expansion

$$p_k^{it} = k^{it}(\log k)^{it}\left(1+\frac{\log\log k-1}{\log k}+O\left(\frac{(\log\log k)^2}{(\log k)^2}\right)\right)^{it} \tag{4.2}$$

$$= k^{it}(\log k)^{it}\left(1+it\frac{\log\log k-1}{\log k}+O_t\left(\frac{(\log\log k)^2}{(\log k)^2}\right)\right). \tag{4.3}$$

Heuristic 4.1 (Principal zeta model)**.** This is a heuristic argument. Retaining only the leading term in (4.3), the first component of (2.7) becomes

$$\sum_{k\geq 1}\mu(k)p_k^{it}k^{-s}\approx\sum_{k\geq 1}\mu(k)(\log k)^{it}k^{-(s-it)}. \tag{4.4}$$

Similarly,

$$\sum_{k\geq 1}\mu(k)p_k^{-it}k^{-s}\approx\sum_{k\geq 1}\mu(k)(\log k)^{-it}k^{-(s+it)}. \tag{4.5}$$

If

$$L_\tau(z)=\sum_{k\geq 1}\mu(k)(\log k)^{i\tau}k^{-z}, \tag{4.6}$$

then the leading model is

$$D_t^{(0)}(s)=\frac{c^{it}}{2i}L_t(s-it)-\frac{c^{-it}}{2i}L_{-t}(s+it). \tag{4.7}$$

Formally, since multiplication by $(\log k)^m$ corresponds to $(-\partial_z)^m$ on a Dirichlet series, the non-integral logarithmic weight suggests

$$L_\tau(z)\sim(-\partial_z)^{i\tau}\frac{1}{\zeta(z)}. \tag{4.8}$$

The lower-order terms in (4.3) produce additional series containing factors of $(\log k)^{it-m}(\log\log k)^r$. They are not discarded analytically; in the reduction below they are part of the singularity-inheritance problem.

5. The shifted singularity calculation

The following computation is formal but decisive for locating the possible singularities of the principal model.

Heuristic 5.1 (Local singularity produced by a simple zero)**.** Let ρ be a simple zero of ζ. Near $z=\rho$ one has

$$\frac{1}{\zeta(z)}=\frac{1}{\zeta'(\rho)(z-\rho)}+O(1). \tag{5.1}$$

Applying the formal fractional logarithmic differential operator gives

$$(-\partial_z)^{it}\frac{1}{\zeta(z)}\sim\frac{\Gamma(1+it)}{\zeta'(\rho)}(z-\rho)^{-1-it}. \tag{5.2}$$

Likewise,

$$(-\partial_z)^{-it}\frac{1}{\zeta(z)} \sim \frac{\Gamma(1-it)}{\zeta'(\rho)}(z-\rho)^{-1+it}. \tag{5.3}$$

Substitution into (4.7) gives expected singularities of $D_t^{(0)}(s)$ at

$$s = \rho + it \tag{5.4}$$

and

$$s = \rho - it. \tag{5.5}$$

The corresponding principal coefficients are

$$C_+(\rho,t) = \frac{c^{it}\Gamma(1+it)}{2i\,\zeta'(\rho)}, \tag{5.6}$$

and

$$C_-(\rho,t) = -\frac{c^{-it}\Gamma(1-it)}{2i\,\zeta'(\rho)}. \tag{5.7}$$

For $t \neq 0$ and simple ρ, neither coefficient vanishes in the principal model.

The real parts of the shifted singularities are unchanged:

$$\Re(\rho + it) = \Re(\rho - it) = \Re\rho. \tag{5.8}$$

Thus any non-removable singularity inherited from a zero with $\Re\rho > 1/2$ lies in the same right half-plane.

6. Singularity transfer for the exact prime-index kernel

The principal model in Section 5 shows where the shifted singularities must occur. This section records the local transfer calculation that prevents the exact prime-index factor p_k^{it} from cancelling the leading inherited branch.

Definition 6.1 (One-sided complex Dirichlet components)**.** For $t \in \mathbb{R}$ set

$$F_{t,+}(s) = \sum_{k\geq 1} \mu(k) p_k^{it} k^{-s}, \tag{6.1}$$

and

$$F_{t,-}(s) = \sum_{k\geq 1} \mu(k) p_k^{-it} k^{-s}. \tag{6.2}$$

Then

$$D_t(s) = \frac{c^{it}}{2i}F_{t,+}(s) - \frac{c^{-it}}{2i}F_{t,-}(s). \tag{6.3}$$

Theorem 6.2 (Conditional fractional logarithmic transfer)**.** *Let*

$$F(z) = \sum_{n\geq 1} a_n n^{-z} \tag{6.4}$$

be a Dirichlet series. Suppose that, near a point $z = \rho$ and on a fixed branch of $z - \rho$, it has a local singular expansion

$$F(z) = A(z-\rho)^{-m} + G(z), \qquad A \neq 0, \tag{6.5}$$

where m is a positive integer and G is less singular at ρ. For $\lambda \in \mathbb{C}$ with

$$m + \lambda \notin \{0, -1, -2, \ldots\}, \tag{6.6}$$

assume that the logarithmic transform

$$\mathcal{L}_\lambda F(z) = \sum_{n\geq 1} a_n (\log n)^\lambda n^{-z} \tag{6.7}$$

admits the local continuation obtained by analytic continuation in λ from the integer identities

$$\mathcal{L}_r F(z) = (-\partial_z)^r F(z), \qquad r = 0, 1, 2, \dots. \tag{6.8}$$

Then the principal local term is

$$\mathcal{L}_\lambda F(z) = A\frac{\Gamma(m+\lambda)}{\Gamma(m)}(z-\rho)^{-m-\lambda} + \text{less singular terms}. \tag{6.9}$$

In particular, if ρ is a simple zero of ζ, then

$$\mathcal{L}_\lambda\left(\frac{1}{\zeta}\right)(z) = \frac{\Gamma(1+\lambda)}{\zeta'(\rho)}(z-\rho)^{-1-\lambda} + \text{less singular terms} \tag{6.10}$$

locally at $z = \rho$.

For comparison, if instead a function has a local zero-type term

$$H(z) = B(z-\rho)^m + \text{less singular terms}, \qquad m \in \mathbb{N}, \tag{6.11}$$

and the same branch convention is used, analytic continuation of $(-\partial_z)^r$ gives

$$(-\partial_z)^\lambda H(z) = B(-1)^\lambda \frac{\Gamma(m+1)}{\Gamma(m-\lambda+1)}(z-\rho)^{m-\lambda} + \text{less singular terms}. \tag{6.12}$$

Thus the factor $\Gamma(1+\lambda)$ in (6.10) belongs to the reciprocal-zeta pole, not to the zero of ζ itself.

Proof. For an integer $r \geq 0$, multiplication of the n-th Dirichlet coefficient by $(\log n)^r$ is the operation $(-\partial_z)^r$ on the series. Applied to the pole term in (6.5), this gives

$$(-\partial_z)^r A(z-\rho)^{-m} = A\frac{\Gamma(m+r)}{\Gamma(m)}(z-\rho)^{-m-r}. \tag{6.13}$$

The right-hand side is meromorphic in r. The conditional definition of $\mathcal{L}_\lambda$ is precisely the continuation of this local identity from nonnegative integers to complex λ, with the branch of $z-\rho$ fixed. This proves (6.9).

If ρ is a simple zero of ζ, then

$$\frac{1}{\zeta(z)} = \frac{1}{\zeta'(\rho)}(z-\rho)^{-1} + O(1). \tag{6.14}$$

Taking $m = 1$ and $A = 1/\zeta'(\rho)$ in (6.9) gives (6.10).

For the zero-type comparison, one computes for integer $r \geq 0$

$$(-\partial_z)^r B(z-\rho)^m = (-1)^r B\frac{\Gamma(m+1)}{\Gamma(m-r+1)}(z-\rho)^{m-r}. \tag{6.15}$$

The factor $(-1)^\lambda$ denotes the branch continuation of the integer factor $(-1)^r$. Continuing the coefficient in r gives (6.12). This also shows why applying the fractional logarithmic transform to a pole of $1/\zeta$ and applying a fractional derivative to a zero of ζ are not the same local calculation. □

Corollary 6.3 (Weighted reciprocal-zeta singularity)**.** *Let ρ be a simple zero of ζ. For*

$$L_\lambda(z) = \sum_{n\geq 1} \mu(n)(\log n)^\lambda n^{-z}, \tag{6.16}$$

the local singular expansion at $z = \rho$ has principal term

$$L_\lambda(z) = \frac{\Gamma(1+\lambda)}{\zeta'(\rho)}(z-\rho)^{-1-\lambda} + \textit{less singular terms}. \tag{6.17}$$

Proof. Since

$$\sum_{n\geq 1}\mu(n)n^{-z}=\frac{1}{\zeta(z)}, \tag{6.18}$$

and ρ is simple,

$$\frac{1}{\zeta(z)}=\frac{1}{\zeta'(\rho)}(z-\rho)^{-1}+O(1). \tag{6.19}$$

Apply Theorem 6.2 with $m=1$ and $A=1/\zeta'(\rho)$. □

The expansion of p_k has the Cipolla form

$$p_k=k\log k\left(1+\sum_{m=1}^{M}\frac{P_m(\log\log k)}{(\log k)^m}+O_M\left(\frac{(\log\log k)^{M+1}}{(\log k)^{M+1}}\right)\right), \tag{6.20}$$

where each P_m is a polynomial; see [1, 2]. For fixed t, raising (6.20) to it gives

$$p_k^{it}=k^{it}(\log k)^{it}\left(1+\sum_{m=1}^{M}\frac{Q_m(t,\log\log k)}{(\log k)^m}+O_{M,t}\left(\frac{(\log\log k)^{M+1}}{(\log k)^{M+1}}\right)\right), \tag{6.21}$$

where $Q_m(t,\cdot)$ is a polynomial.

Lemma 6.4 (Lower order of the prime-index corrections)**.** *Let ρ be a simple zero of ζ and fix $t\neq 0$. The leading term of $F_{t,+}$ in* (6.1) *has at $s=\rho+it$ the branch*

$$\frac{\Gamma(1+it)}{\zeta'(\rho)}(s-it-\rho)^{-1-it}. \tag{6.22}$$

Every correction term arising from the m-th prime-index correction in (6.21)*, with $m\geq 1$, has local singular order at most*

$$(s-it-\rho)^{-1-it+m} \tag{6.23}$$

up to powers of $\log(s-it-\rho)$. Consequently no correction term with $m\geq 1$ can cancel the leading branch (6.22)*.*

Proof. The leading term in (6.21) contributes

$$\sum_{k\geq 1}\mu(k)(\log k)^{it}k^{-(s-it)}=L_{it}(s-it). \tag{6.24}$$

By Corollary 6.3, this gives (6.22).

A correction term of order $m\geq 1$ is a finite linear combination of series of the form

$$\sum_{k\geq 1}\mu(k)(\log k)^{it-m}(\log\log k)^r k^{-(s-it)}, \tag{6.25}$$

with $0\leq r\leq \deg Q_m$. Ignoring the factor $(\log\log k)^r$ for a moment, Corollary 6.3 gives singular order

$$(s-it-\rho)^{-1-(it-m)}=(s-it-\rho)^{-1-it+m}. \tag{6.26}$$

The additional factor $(\log\log k)^r$ corresponds locally to r differentiations with respect to the logarithmic exponent parameter; therefore it can introduce powers of $\log(s-it-\rho)$ but it does not restore the missing integer power m in (6.26). Hence every $m\geq 1$ term is strictly less singular than (6.22) and cannot cancel it. □

Theorem 6.5 (Non-removable inheritance at a simple zero)**.** *Let $t \neq 0$, $c > 0$, and let ρ be a simple nontrivial zero of ζ. In the local singular expansion generated by the prime-index asymptotic transfer, the Dirichlet series $D_t(s)$ has non-removable singular branches at*

$$s = \rho + it \tag{6.27}$$

and

$$s = \rho - it. \tag{6.28}$$

The corresponding leading branches are

$$D_t(s) = \frac{c^{it}\Gamma(1+it)}{2i\,\zeta'(\rho)}(s - it - \rho)^{-1-it} + \text{less singular terms} \tag{6.29}$$

near $s = \rho + it$, and

$$D_t(s) = -\frac{c^{-it}\Gamma(1-it)}{2i\,\zeta'(\rho)}(s + it - \rho)^{-1+it} + \text{less singular terms} \tag{6.30}$$

near $s = \rho - it$. In particular the leading inherited coefficients are nonzero.

Proof. The branch (6.29) follows from the plus component in (6.3), Lemma 6.4, and multiplication by the nonzero factor $c^{it}/(2i)$. The same argument applied to p_k^{-it} gives the branch (6.30); the exponent changes from $-1-it$ to $-1+it$, and the nonzero prefactor is $-c^{-it}/(2i)$. Since $\Gamma(1+it)$, $\Gamma(1-it)$, c^{it}, c^{-it}, and $\zeta'(\rho)$ are nonzero for a simple zero, the coefficients in (6.29) and (6.30) are nonzero. Lower prime-index corrections have strictly smaller singular order by Lemma 6.4. They cannot cancel the leading branches. □

Hypothesis 6.6 (Strong fractional transfer)**.** *Assume that the local fractional logarithmic transfer principle of Theorem 6.2 extends to the complete prime-index logarithmic symbol generated by the Cipolla expansion. Thus, for a simple zero ρ of ζ, the leading factor $k^{it}(\log k)^{it}$ produces the inherited branch described in Theorem 6.5; every explicit Cipolla correction has strictly lower local singular order; and the exact asymptotic remainder is transferred to terms strictly less singular than that leading branch. In particular, the prime-index corrections and their remainder do not cancel the inherited leading branch.*

Hypothesis 6.7 (Square-root bound for the odd kernel)**.** *For a fixed nonzero frequency t and for every $\varepsilon > 0$, the odd log-prime kernel satisfies*

$$Y_t(X) = O_{t,c,\varepsilon}(X^{1/2+\varepsilon}). \tag{6.31}$$

Theorem 6.8 (Conditional link to the Riemann Hypothesis)**.** *Assume Hypotheses 6.6 and 6.7. Then $\zeta(s)$ has no simple nontrivial zero with $\Re\rho > 1/2$. Consequently, under these hypotheses, the Riemann Hypothesis holds for simple zeros.*

Proof. By Hypothesis 6.7 and Corollary 3.2, $D_t(s)$ is holomorphic in the half-plane $\Re s > 1/2$. Let ρ be a simple nontrivial zero of ζ. By Hypothesis 6.6 and Theorem 6.5, the zero ρ produces a non-removable inherited singular branch of $D_t(s)$ at one of the shifted points $s = \rho \pm it$, with

$$\Re(\rho \pm it) = \Re\rho. \tag{6.32}$$

If $\Re\rho > 1/2$, this singular point lies in the half-plane where $D_t(s)$ is holomorphic, a contradiction. Hence no simple zero can lie in $\Re s > 1/2$. The functional equation reflects nontrivial zeros across the critical line, so the absence of simple zeros on the right of the critical line implies that every simple nontrivial zero lies on $\Re s = 1/2$. □

7. Relation with the Odlyzko–te Riele mechanism

The Odlyzko–te Riele argument starts from the classical identity (1.5). Mertens-type square-root control of $M(X)$ would imply analytic continuation of $1/\zeta(s)$ into $\Re s > 1/2$, and the explicit formula then expresses the normalized Mertens function in terms of zero phases and coefficients involving $1/(\rho\zeta'(\rho))$ [7]. The computational step searches for large values by aligning phases through simultaneous Diophantine approximation.

For the present kernel, the analogous normalized variable is

$$u = \log X, \qquad \mathcal{Y}_t(u) = e^{-u/2} Y_t(e^u). \tag{7.1}$$

A smoothed explicit formula suggested by the preceding singularity calculation has the schematic form

$$\mathcal{Y}_{t,h}(u) = \Im \sum_{\rho} B_+(\rho, t, h) e^{i(\gamma+t)u} + \Im \sum_{\rho} B_-(\rho, t, h) e^{i(\gamma-t)u} + E_{t,h}(u), \tag{7.2}$$

where h is a smoothing kernel in the u variable and the coefficients contain the transformed weight generated by $p_k^{\pm it}$. The principal model suggested by (5.6) and (5.7) is

$$B_+(\rho, t, h) \approx \frac{c^{it}\Gamma(1+it)\widehat{h}(\gamma+t)}{2i\,\zeta'(\rho)}, \tag{7.3}$$

and

$$B_-(\rho, t, h) \approx -\frac{c^{-it}\Gamma(1-it)\widehat{h}(\gamma-t)}{2i\,\zeta'(\rho)}. \tag{7.4}$$

The finite zero block associated with (7.2) is

$$P_{t,T}(u) = \Im \sum_{0<\gamma\le T} B_+(\rho, t, h) e^{i(\gamma+t)u} + \Im \sum_{0<\gamma\le T} B_-(\rho, t, h) e^{i(\gamma-t)u}. \tag{7.5}$$

This is the object to which the Odlyzko–te Riele lattice method applies. The phase conditions for constructive alignment are

$$(\gamma_j + t)u + \arg B_+(\rho_j, t, h) \approx \frac{\pi}{2} \pmod{2\pi}, \tag{7.6}$$

and

$$(\gamma_j - t)u + \arg B_-(\rho_j, t, h) \approx \frac{\pi}{2} \pmod{2\pi}. \tag{7.7}$$

The shift $\gamma \mapsto \gamma \pm t$ alone does not create suppression: if all coefficients are unchanged, the $+t$ shift is only multiplication by e^{itu}. The effective arithmetic content is therefore in the coefficients (7.3) and (7.4), together with the inherited coefficients in Theorem 6.5.

8. Numerical evidence for the odd kernel

The computation used $c = \pi/2$, primes $p_k \le 2,000,000$, hence

$$K = 148933, \tag{8.1}$$

and the values of $\mu(k)$ for $1 \le k \le K$. The counts were

$$\#\{k : \mu(k) = -1\} = 45298, \quad \#\{k : \mu(k) = 0\} = 58389, \quad \#\{k : \mu(k) = 1\} = 45246. \tag{8.2}$$

The null model consisted of random sign permutations preserving the zero positions of $\mu(k)$ and preserving the total number of $+1$ and -1 values.

Experiment 8.1 (Representative odd sine suppression)**.** The statistic measured was

$$\frac{Y_t(K)}{\sqrt{K}} = \frac{1}{\sqrt{K}} \sum_{k \le K} \mu(k) \sin(t \log(cp_k)). \tag{8.3}$$

For each tail $K \ge K_{\min}$ the real metric was divided by the median null metric. Ratios below 1 indicate suppression relative to the sign-permutation ensemble.

$K_{\min}$	ratio std	ratio q90	ratio q95	ratio max	ratio endpoint
100	0.51915	0.36179	0.35534	0.37152	0.33665
1000	0.52099	0.36557	0.35629	0.38079	0.33665
10000	0.57458	0.36846	0.37816	0.46182	0.33665
50000	0.84703	0.43251	0.44925	0.50518	0.33665
100000	1.18960	0.48710	0.49664	0.54739	0.33665

TABLE 1. Representative odd-sine suppression ratios for the initial computation.

The quantile and endpoint statistics are suppressed across all listed large tails. The standard deviation ceases to be suppressed in the last tail, which indicates that robust quantiles and endpoints are more stable than the standard deviation for this kernel.

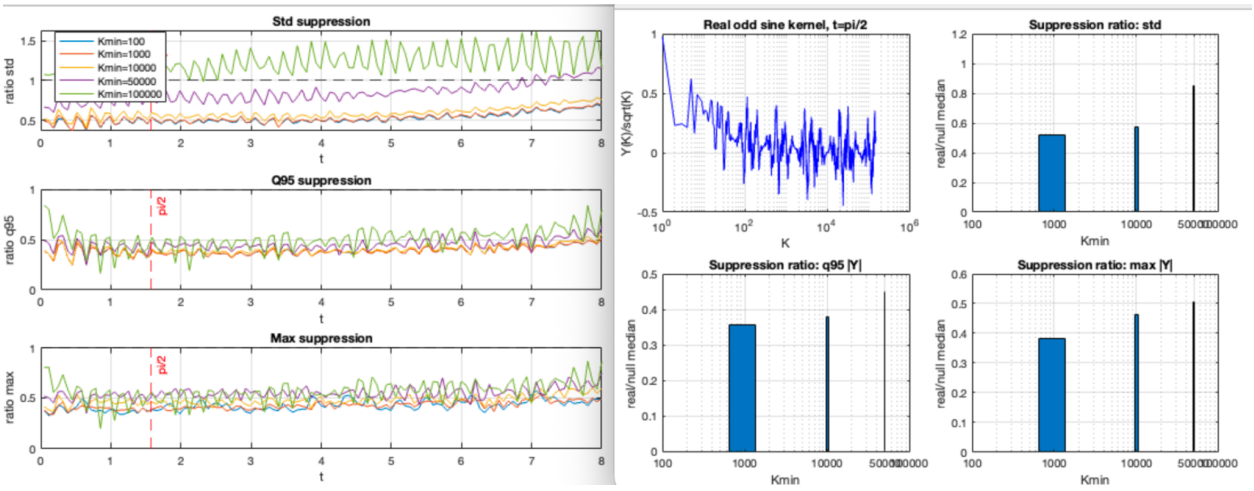


FIGURE 1. Odd sine suppression experiment for $c = \pi/2$. Left: sweep in t for tail statistics. Right: one representative normalized path and tail suppression ratios.

The displayed frequency was chosen before the extended sweep and has no special numerical status. The sweep is used to motivate the analytic study of $D_t(s)$ for general nonzero t.

9. Classical statistics and variance normalization

The sign-permutation experiment in Section 8 compares the real Moebius signs with random signs preserving the squarefree mask. A separate test is needed to check whether the effect is only caused by the marginal distribution of the deterministic weight

$$f_t(k) = \sin(t \log(cp_k)). \tag{9.1}$$

The useful comparison is between the unweighted average of f_t, the squarefree-restricted average of $\mu(k)^2 f_t(k)$, and the signed average of $\mu(k) f_t(k)$.

Definition 9.1 (Variance-normalized kernel statistic)**.** For $1 \le k \le K$, define

$$Y_t(K) = \sum_{j \le K} \mu(j) f_t(j), \tag{9.2}$$

$$V_t(K) = \sum_{j \le K} \mu(j)^2 f_t(j)^2, \tag{9.3}$$

and

$$Z_t(K) = \frac{Y_t(K)}{\sqrt{V_t(K)}}. \tag{9.4}$$

The statistic $Z_t(K)$ measures the signed sum against the quadratic scale generated by its own weight.

The computation was extended to primes

$$p_k \le 2.5 \cdot 10^8, \qquad K = 13,679,318. \tag{9.5}$$

The corresponding Moebius counts were

$$\#\{\mu = -1\} = 4,158,169, \quad \#\{\mu = 0\} = 5,363,273, \quad \#\{\mu = 1\} = 4,157,876. \tag{9.6}$$

One non-optimized frequency was retained as an arbitrary reference cross-section of the two-parameter data. It was not selected because of an extremal statistic; the extended sweep shows stronger low-escape frequencies elsewhere.

Experiment 9.2 (Extended classical statistics)**.** At the representative frequency, the extended computation gives

$$\frac{1}{K} \sum_{k \le K} f_t(k) = -0.50352, \qquad \frac{1}{K} \sum_{k \le K} \mu(k)^2 f_t(k) = -0.30610, \tag{9.7}$$

while the signed average is

$$\frac{1}{K} \sum_{k \le K} \mu(k) f_t(k) = 1.3289 \cdot 10^{-5}. \tag{9.8}$$

The second moment and endpoint normalization are

$$\frac{1}{K} \sum_{k \le K} \mu(k)^2 f_t(k)^2 = 0.33571, \qquad \sqrt{V_t(K)} = 2143, \tag{9.9}$$

and

$$Y_t(K) = 181.79, \qquad \frac{Y_t(K)}{\sqrt{K}} = 0.049151, \qquad Z_t(K) = 0.08483. \tag{9.10}$$

The first two averages in (9.7) are not small. Thus the weight is not being centered by itself, nor by the squarefree mask. The signed average in (9.8) is small only after multiplication by the Moebius signs. This is the basic statistic excluding the trivial explanation by marginal equidistribution of $f_t(k)$.

Experiment 9.3 (Extended tail compression)**.** For the same representative frequency, the tail quantiles remain stable when the initial segment is removed. The columns $q95_{\sqrt{K}}$ and $\max_{\sqrt{K}}$ refer to $|Y_t(K)|/\sqrt{K}$ on $K \ge K_{\min}$, while $q95_Z$ and $\max_Z$ refer to $|Z_t(K)|$ on the same tail.

The quantile statistics in Table 2 are essentially unchanged from $K_{\min} = 100$ to $K_{\min} = 100000$. This shows that the compression is not an artifact of the first few terms. The maximum is larger and more sensitive, as expected for an extremal statistic, but it also decreases after removing the initial segment.

$K_{\min}$	$q95_{\sqrt{K}}$	$\max_{\sqrt{K}}$	$q95_Z$	$\max_Z$
100	0.26111	0.47049	0.47445	0.83065
1000	0.26111	0.44661	0.47443	0.78534
10000	0.26112	0.44661	0.47445	0.78534
100000	0.26114	0.42119	0.47437	0.73218

TABLE 2. Extended tail statistics for the odd log-prime kernel after square-root and variance normalization.

Proposition 9.4 (Numerical interpretation of the extended variance test)**.** *In the extended computation, the odd log-prime kernel is not suppressed merely because $f_t(k)$ has a small marginal mean or variance. The suppression persists after normalization by $V_t(K)$ in* (9.3)*, and it remains stable under tail truncation.*

Proof. This is a numerical proposition based on Experiments 9.2 and 9.3. The marginal averages in (9.7) are not close to zero, whereas the signed average (9.8) is much smaller. The endpoint value in (9.10) gives $Z_t(K) = 0.08483$, and the tail values in Table 2 give $q95_Z \simeq 0.474$ across the displayed truncations. Hence the smallness survives the intrinsic variance normalization and is not explained by the weight alone. □

Experiment 9.5 (Random-sign trajectory envelopes)**.** The previous statistics compress each frequency to a small number of tail functionals. To test whether the effect is visible at the level of the whole path, three frequencies were then compared with random-sign walks on the same squarefree support. The null paths were

$$Y_t^{(r)}(K) = \sum_{j \le K} \varepsilon_j^{(r)} \mu(j)^2 \sin\big(t \log(cp_j)\big), \tag{9.11}$$

where the signs $\varepsilon_j^{(r)} \in \{-1, 1\}$ are independent on the squarefree support. The experiment used 200 null paths and 2231 logarithmically spaced sample points up to $K = 13,679,318$.

The extended sweep ranks $t = 3.0437$, $t = 0.48243$, and $t = 7.8337$ among the strongest low-escape frequencies in the tested range. Detailed trajectory plots were generated for $t = 3.0437$, $t = 0.48243$, and one arbitrary non-optimized reference frequency. Table 3 reports the endpoint statistic and the comparison with the null envelopes for the plotted paths. The columns $q95_Z$ and $\max_Z$ are computed on the tail $K \ge 10^5$; the null columns give the median of the same statistic over the null paths. The quantity $p_{\le}$ is the fraction of null realizations with statistic less than or equal to the real statistic.

t	$Z_t(K)$	below null median	real $q95_Z$	null median $q95_Z$	real $\max_Z$	null median $\max_Z$
3.0437	−0.001415	0.991484	0.43850	1.5953	0.78673	1.9732
0.48243	0.004730	0.981623	0.27038	1.5965	0.71139	1.9996
1.5708 (arb.)	0.084830	0.994621	0.47437	1.5919	0.73218	2.0142

TABLE 3. Trajectory-level comparison with random-sign null paths on the same squarefree support. In all three displayed cases, $p_{\le} = 0$ for both $q95_Z$ and $\max_Z$ at $K_{\min} = 10^5$ among the 200 null realizations.

The table shows that the real paths are not merely ending close to zero. They remain below the median absolute null envelope on more than 98% of the sampled points, and the tail $q95_Z$ and $\max_Z$ statistics are below every sampled null realization at $K_{\min} = 10^5$ for the three displayed frequencies. Thus the effect is trajectory-level compression relative to the natural random-sign model, not just endpoint cancellation.

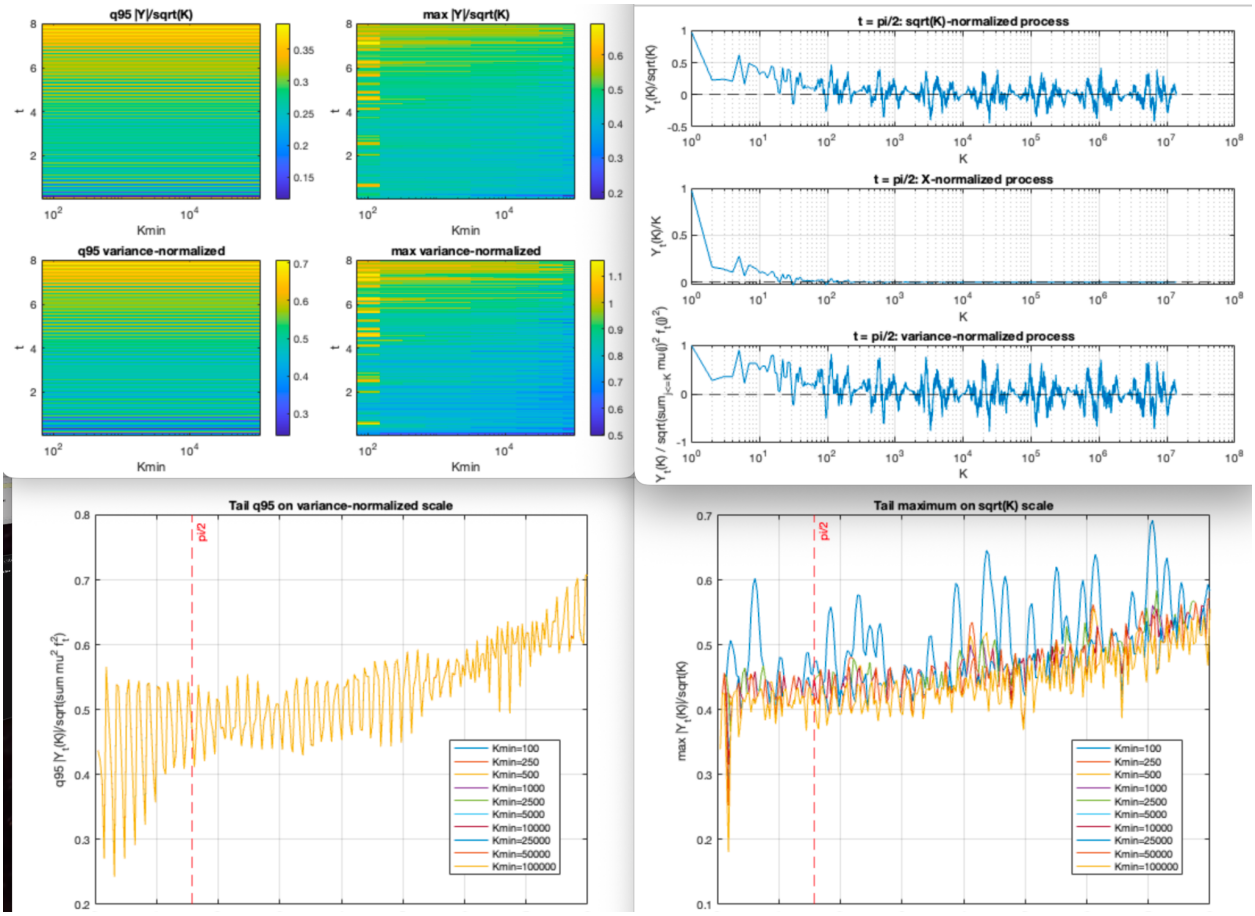


FIGURE 2. Extended classical statistics for the odd log-prime Moebius kernel up to $p_k \le 2.5 \cdot 10^8$. The dashboard shows heatmaps for tail quantiles and maxima, together with one representative normalized path. The variance-normalized quantiles remain compressed under tail truncation.

The extended sweep also provides a direct variance-scale summary for the three selected low-escape frequencies. Table 4 uses $K = 13,679,318$ and reports the endpoint value in intrinsic standard deviation units, together with tail statistics at $K_{\min} = 10^5$. For $t = 3.0437$ and $t = 0.48243$, the random-sign trajectory test also gives the median null value of the same $q95_Z$ statistic. The value $t = 7.8337$ is the third selected low-escape frequency from the sweep; it was not included in the trajectory-envelope plots, so only its sweep-level variance statistics are reported.

t	$\lvert Z_t(K)\rvert$	$q95_Z$	$\max_Z$	null median $q95_Z$	$q95_Z$/null median
3.0437	0.001415	0.43850	0.78673	1.5953	0.275
0.48243	0.004730	0.27038	0.71139	1.5965	0.169
7.8337	0.002147	0.70299	0.91811	—	—

TABLE 4. Variance-scale cancellation for the three selected low-escape frequencies from the extended sweep. The endpoint statistic $|Z_t(K)|$ measures the final cancellation in units of the internal standard deviation. The tail statistics are computed on $K \ge 10^5$.

Proposition 9.6 (Numerical interpretation of the trajectory null test)**.** *In the extended computation, the low-escape behavior is visible as a compressed trajectory relative to random signs on the*

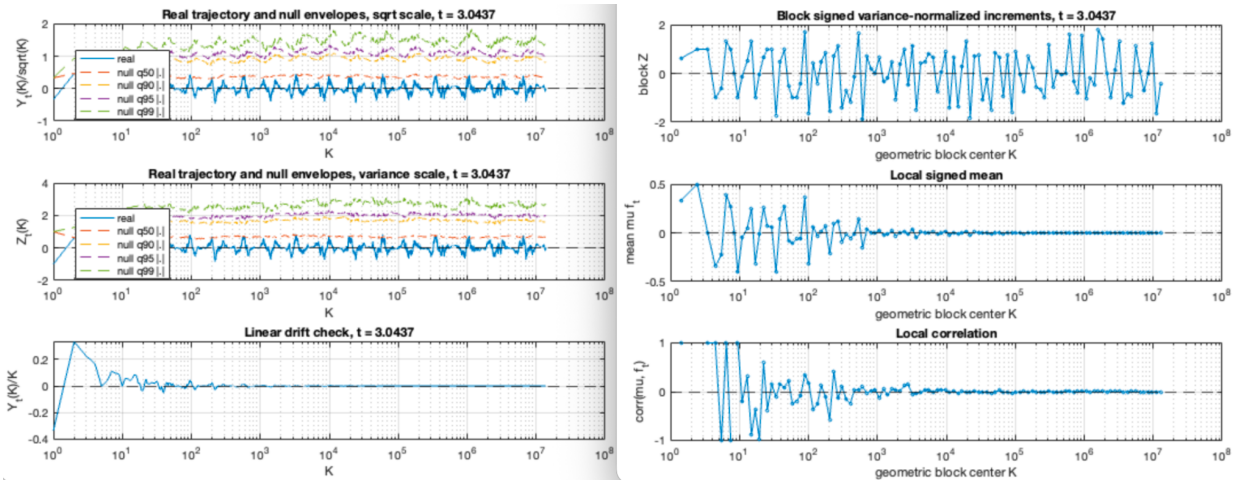


FIGURE 3. Trajectory and local block diagnostics for the low-escape frequency $t = 3.0437$. Left: real path compared with random-sign absolute null envelopes on the $\sqrt{K}$ scale and on the intrinsic variance scale. Right: logarithmic block diagnostics. The path remains below the typical null envelopes even though the signed block increments fluctuate on a normal local scale.

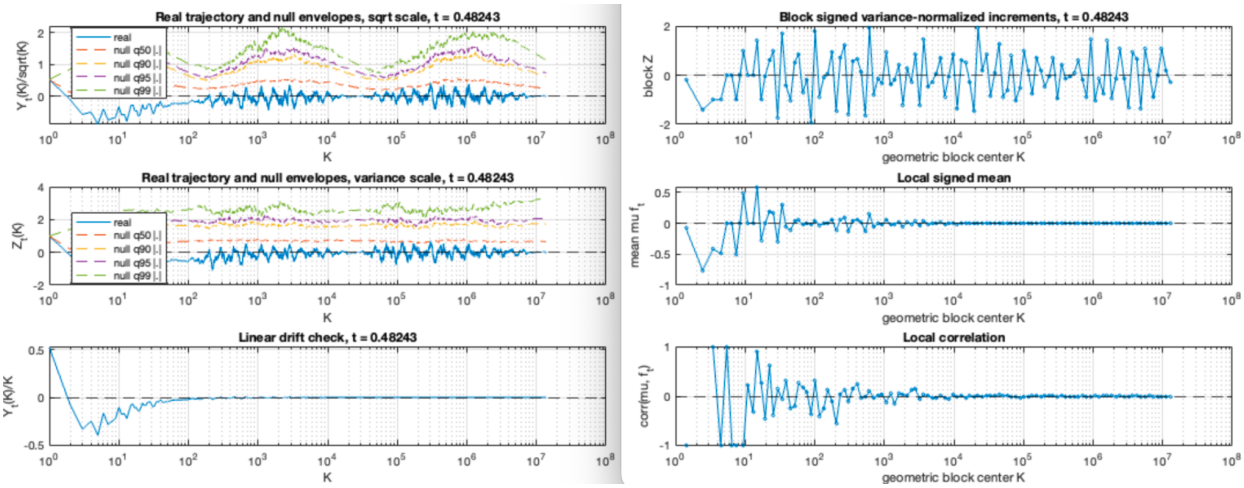


FIGURE 4. Trajectory and local block diagnostics for the low-escape frequency $t = 0.48243$. The real trajectory is again compressed relative to random-sign walks with the same support and deterministic weight. The linear drift diagnostic tends rapidly to zero.

same squarefree support. It is not only an endpoint cancellation and not a consequence of uniformly small local blocks.

Proof. The endpoint values alone would not distinguish a persistent low-escape path from a late accidental crossing. Experiment 9.5 compares the whole sampled path with the envelopes generated by (9.11). For $t = 3.0437$, 0.48243, and the arbitrary reference frequency, the real path is below the null median on respectively 0.991484, 0.981623, and 0.994621 of the sample grid. At $K_{\min} = 10^5$, both the tail quantile $q95_Z$ and the tail maximum $\max_Z$ have $p_{\le} = 0$ among 200 null realizations for all three displayed frequencies. Hence the compression is seen throughout the path and in the tail functionals, not only at the final point. The block plots in Figures 3– 5 show signed block increments of order one; therefore the observed effect should be interpreted as unusually small

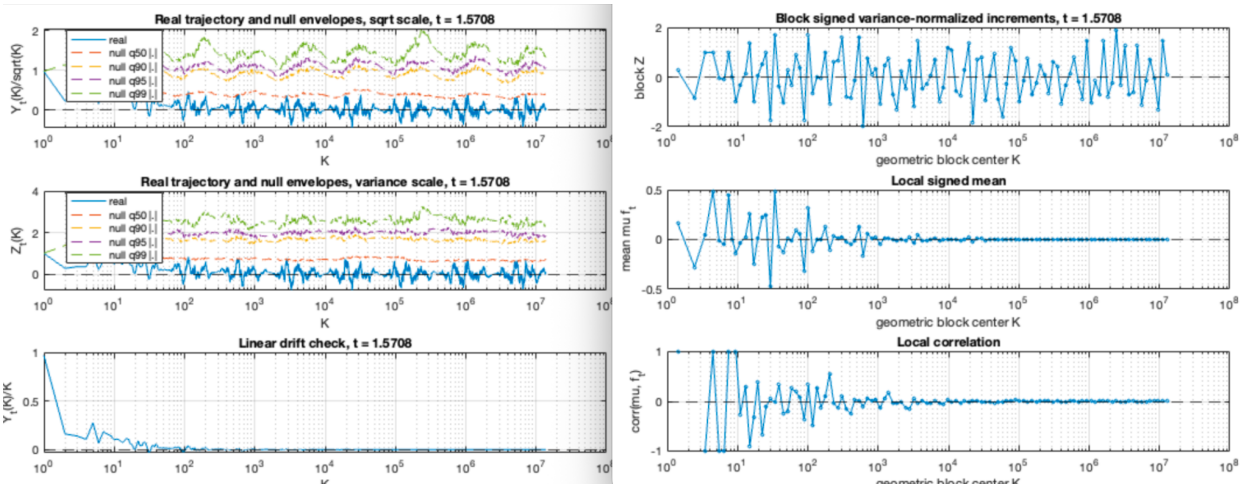


FIGURE 5. Trajectory and local block diagnostics for the non-optimized arbitrary reference frequency $t = 1.5708$. Its role is only to show that the compression is not confined to frequencies selected by an extremal sweep.

long-range escape of ordinary local fluctuations, rather than as uniform local annihilation block by block. □

The statistical conclusion is that the observed smallness is a property of the signed product

$$\mu(k)\sin\big(t\log(cp_k)\big), \tag{9.12}$$

not of the deterministic factor alone. The trajectory-level null test strengthens this conclusion: the real Moebius signs produce paths with substantially less escape than random signs on the same squarefree support. This supports the use of the Dirichlet series $D_t(s)$ in (2.6) as the analytic object attached to the kernel.

10. Conclusion

The main purpose of this work has been to exhibit finite-range cancellation for the odd log-prime kernel

$$Y_t(X) = \sum_{k\le X} \mu(k)\sin(t\log(cp_k)) \tag{10.1}$$

and to isolate the analytic conditions under which this cancellation would extend to an asymptotic regime. The kernel leads to the Dirichlet series

$$D_t(s) = \sum_{k\ge 1} \frac{\mu(k)\sin(t\log(cp_k))}{k^s}. \tag{10.2}$$

A square-root scale estimate for (10.1) implies, by Abelian continuation, holomorphy of (10.2) beyond the critical boundary.

The local singularity transfer carried out in Section 6 shows how the exact prime-index factor $p_k^{\pm it}$ inherits the leading singular branches of the shifted reciprocal zeta factors. For a simple zero ρ of ζ, the inherited branches occur at

$$s = \rho + it \tag{10.3}$$

and

$$s = \rho - it, \tag{10.4}$$

with leading coefficients

$$\frac{c^{it}\Gamma(1+it)}{2i\,\zeta'(\rho)} \tag{10.5}$$

and

$$-\frac{c^{-it}\Gamma(1-it)}{2i\,\zeta'(\rho)}. \tag{10.6}$$

The explicit Cipolla corrections to p_k have lower local singular order. The strong fractional transfer hypothesis isolates the remaining analytic requirement: the exact logarithmic remainder of the prime-index symbol must also be locally lower order and must not reconstruct the leading branch.

The numerical experiments for fixed c give a finite-range test of this mechanism. In the extended sweep through the first 225 evaluated frequencies, the selected low-escape values are

$$t = 3.0437, \qquad t = 0.48243, \qquad t = 7.8337. \tag{10.7}$$

At $p_k \le 2.5 \cdot 10^8$, the corresponding endpoint cancellation on the intrinsic variance scale is

$$|Z_t(K)| = 0.001415, \qquad |Z_t(K)| = 0.004730, \qquad |Z_t(K)| = 0.002147, \tag{10.8}$$

respectively. The detailed random-sign trajectory tests for the first two selected frequencies show that the real paths remain compressed relative to null walks on the same squarefree support and with the same deterministic weight. The effect is therefore not only endpoint cancellation and is not explained by the marginal distribution of the phase weight alone.

The strong fractional transfer hypothesis that emerges from this analysis does not follow, at least trivially, from the Riemann Hypothesis. RH imposes global restrictions on the location of zeros, but it does not by itself guarantee the local analytic stability, symbolic logarithmic transfer, or microlocal non-cancellation required by the mechanism above. Consequently, even if RH is true, the analytic condition identified here retains separate content.

Under the strong transfer hypothesis and a square-root bound for $Y_t(X)$, Theorem 6.8 forces every simple nontrivial zero of $\zeta(s)$ onto the critical line. Thus the paper reduces the conditional route to two explicit requirements: local symbolic transfer for the exact prime-index logarithmic factor, and square-root cancellation for the odd log-prime kernel.

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